

Appu Gowda H C

appugowda558@gmail.com / 9164774147 / Bengaluru, India

SUMMARY

A proactive and adaptable software engineer with a strong foundation in coding and problem-solving, I am passionate about tackling complex challenges and continuously improving my skill set. With extensive experience in collaborative projects, I thrive in team environments where effective communication and knowledge sharing are key to success. I quickly adapt to new technologies and methodologies, ensuring that I can contribute meaningfully from day one. My ability to bridge technical expertise with a commitment to delivering high-quality, user-centric software solutions allows me to make a significant impact on any project I undertake. My background includes proficiency in Algorithm and Data Structure design, enabling efficient software Product Engineering and implementation of Business Logic.

EXPERIENCE

Software Engineer	Criterion Networks Private Ltd. • Bengaluru	06/2022 - Present
--------------------------	---	-------------------

AWS Lambda and API Gateway:

- Designed and developed AWS Lambda functions using Python to automate virtual machine provisioning on Google Cloud Platform (GCP), reducing VM creation downtime, and improving efficiency.
- Automated VM lifecycle management, including suspend, resume, and snapshot creation, eliminating manual intervention.
- Enhanced reliability and fault tolerance by implementing failover mechanisms to minimize VM setup failures.
- Optimized resource utilization by integrating event-driven automation, ensuring efficient allocation of compute resources.
- Improved deployment speed and operational efficiency by reducing provisioning time through serverless architecture.

OAuth2 Authentication Integration:

- Implemented OAuth2-based authentication for Google and Facebook, enabling seamless and secure user login.
- Enhanced security and user experience by integrating third-party authentication, reducing the need for password-based logins.
- Ensured compliance with OAuth2 standards, including token validation, user authorization, and secure session management.
- Improved onboarding efficiency by enabling users to authenticate with existing credentials, reducing sign-up friction.

Push notification System:

- Designed and implemented a real-time push notification system to keep users informed about critical events, alerts, successes, failures, and new feature releases.
- Enabled instant user engagement by integrating notifications across web and mobile platforms.
- Ensured reliable message delivery using WebSockets and cloud-based notification services.
- Improved system transparency and user experience by providing immediate feedback on platform activities.

Multiple Virtual Instances with Terraform:

- Experienced challenges related to efficiency, timing, reliability, and resource management while managing all network topology devices on one virtual instance.
- Devised distributed approach utilizing multiple instances for a maximum 300-node network topology.
- Utilized Terraform to efficiently deploy multiple instances simultaneously on Google Cloud.

Automated Network Device Onboarding:

- Developed an end-to-end automated onboarding system to replace the manual process of adding network devices to the platform.
- Designed a user-friendly interface using React (MUI) and implemented backend logic with Django to streamline the upload and configuration of qcow2 files (up to 50GB) on Google Cloud.
- Optimized large file uploads by implementing Google Cloud's resumable upload with chunking, ensuring reliability and efficiency.
- Automated device property configuration and cloud storage integration, eliminating manual errors and inefficiencies.

- Enhanced scalability and deployment speed, enabling seamless onboarding of multiple devices with minimal user intervention.

Microservices Containerization & CI/CD Automation:

- Containerized all microservices using Docker, eliminating package installation failures and reducing service breakdowns.
- Developed and automated a CI/CD pipeline to streamline microservice deployment across staging, beta, and production environments.
- Implemented Bitbucket-based CI/CD automation, significantly reducing manual intervention and deployment errors.
- Optimized build, test, and deployment workflows, ensuring faster, more reliable, and consistent releases.
- Enhanced scalability and operational efficiency, enabling seamless integration and continuous delivery of microservices.

Cluster Monitor tool:

- Developed a Cluster Monitoring Tool to manage overlay network devices hosted on KVM, dynamically allocating CPU, RAM, storage, and network resources using the Python FastAPI Framework.
- Implemented WebSocket communication and Cloud API integration to enable real-time auto-scaling, allowing clusters to scale up or down based on usage demands.
- Designed an interactive dashboard using React (MUI) for real-time visualization of device resources, network performance, and system health.
- Integrated WebSocket-based notifications to instantly alert users about device state changes, resource exhaustion, network failures, and service disruptions.

AWS Resource Automation & Cleanup:

- Automated the creation of AWS resources such as users, roles, policies, VPCs, subnets, gateways, route tables, and VPNs using the AWS Boto3 module, eliminating manual intervention and improving operational efficiency.
- Developed scripts to automate the provisioning of necessary AWS infrastructure, ensuring consistent and error-free resource creation.
- Built a serverless function on Google Cloud Run to automate the cleanup and removal of unused AWS resources, optimizing cost management and resource allocation.
- Enhanced resource provisioning speed and accuracy, streamlining the setup of complex AWS environments.

Self-Initiated Tasks:

- Shrunk AWS RDS database disks across all environments (staging, beta, production) by implementing blue/green deployment strategies to minimize downtime and ensure smooth transitions.
- Optimized 10+ backend APIs by performing query optimizations, enhancing performance, and ensuring only necessary data is served to clients, resulting in faster response times and reduced load on the database through effective Algorithm Design.

Cisco Controller and Router Automated Registration & Configuration:

- Developed an automated registration and configuration system for Cisco controllers and routers using Python multi-threading and multiprocessing, reducing overall provisioning time by 80%.
- Leveraged concurrent execution to parallelize device discovery, credential push, and configuration deployment across large-scale network environments.
- Eliminated manual device setup steps, improving operational consistency and significantly accelerating network onboarding workflows.

Agentic Platform Development & AWS Deployment:

- Developed a full-stack Agentic AI platform from the ground up using React (frontend), FastAPI (backend), and PostgreSQL (database), delivering a scalable and responsive user experience for network operations teams.
- Designed and implemented RESTful APIs with FastAPI to support agent task management, session handling, and real-time status updates, ensuring high performance and low-latency communication between services.
- Architected and deployed the platform on AWS using ECS (Elastic Container Service) with containerized microservices, provisioning an Application Load Balancer for traffic distribution and high availability.
- Configured VPC with public/private subnets, security groups, and IAM roles to enforce network isolation, least-privilege access, and compliance with cloud security best practices.

- Automated infrastructure provisioning and deployment pipelines, enabling consistent and repeatable deployments across staging and production environments.

Email and Webex Notification System:

- Designed and implemented a real-time notification system supporting Email and Cisco Webex channels, enabling instant alerting for network events, agent task completions, and platform incidents.
- Leveraged Redis for fast in-memory message queuing and temporary state storage, ensuring low-latency delivery of high-priority notifications.
- Integrated RabbitMQ as a reliable message broker to decouple notification producers from consumers, guaranteeing message durability, delivery ordering, and retry handling for failed notifications.
- Built configurable notification templates and routing rules, allowing engineers to subscribe to specific event types and receive targeted alerts via their preferred channel.

Agentic AI Platform for SD-WAN and SDA Networks:

- Contributed to the development of an Agentic AI platform tailored for SD-WAN and SDA network environments, enabling network engineers to autonomously diagnose and resolve network issues.
- Implemented AI-driven agents using LangChain and LangGraph to detect network breakages, failures, and degraded paths, providing real-time root cause analysis and remediation suggestions.
- Integrated Temporal workflow orchestration to manage long-running agentic tasks reliably, ensuring fault tolerance and state persistence across complex multi-step network operations.
- Enabled automated configuration push to network devices, reducing manual intervention for routine configuration changes and compliance enforcement across large-scale network topologies.
- Built health monitoring and inventory management capabilities, giving engineers a unified view of device status, interface metrics, and topology across the SD-WAN and SDA fabric.
- Integrated ServiceNow for automated incident creation and ticket lifecycle management, Splunk for log correlation and anomaly detection, and ThousandEyes for end-to-end network path visibility and performance monitoring.
- Leveraged Model Context Protocol (MCP) to standardize tool and data source interactions for AI agents, improving extensibility and interoperability across the platform.

SKILLS

Python, JavaScript, Django, ReactJS, Docker, Terraform, HTML, CSS, JQuery, FastAPI, Redux, Linux, AWS, Google Cloud Platform, Git, GitHub, Jira, Puppet, Bitbucket and Confluence, CI/CD, Agile Methodology, Java, Problem Solving, Team Player, Full Stack development, MySQL, Node.js, MongoDB, Express.js, Object Oriented Design, Object Oriented Programming, Design Patterns, Data Structure and Algorithms, System Design, Collaboration, debugging in linux, LangChain, LangGraph, Temporal, PostgreSQL, Agentic-AI, MCP

EDUCATION

Bachelor Of Engineering

BGS Institute Of Technology • Mandya, Karnataka

10/2020

CGPA = 8.3

Diploma

Government PolyTechnic • Mysore, Karnataka

05/2017

Secondary School

SKES High School • Mandya

05/2014

OTHER DETAILS

[LinkedIn Profile](#)

[GitHub](#)